Embeddings

A Layman's Thought

→ we as a layman can think of representing words in one-hot encoded form.

e.g:

we have words "office", "science", "physics"

their one hot encodings would be:

office : [1, 0, 0]

science : [0, 1, 0]

physics : [0, 0, 1]

Issue:

1) In case of a vocab with 1M words, our OHV would be of size 1M.

2) Here every vector has equal similarity and dissimilarity with others.

i.o.w there is no semantic meaning of embedding.

→ we can also think of using random numbers to represent a word, but still semantic relation-ship of words would not be embedded.

Using Neural Network

→ A simple neural network is used model prediction based on each words:

- predict 'science' and 'physics' based on 'office'.

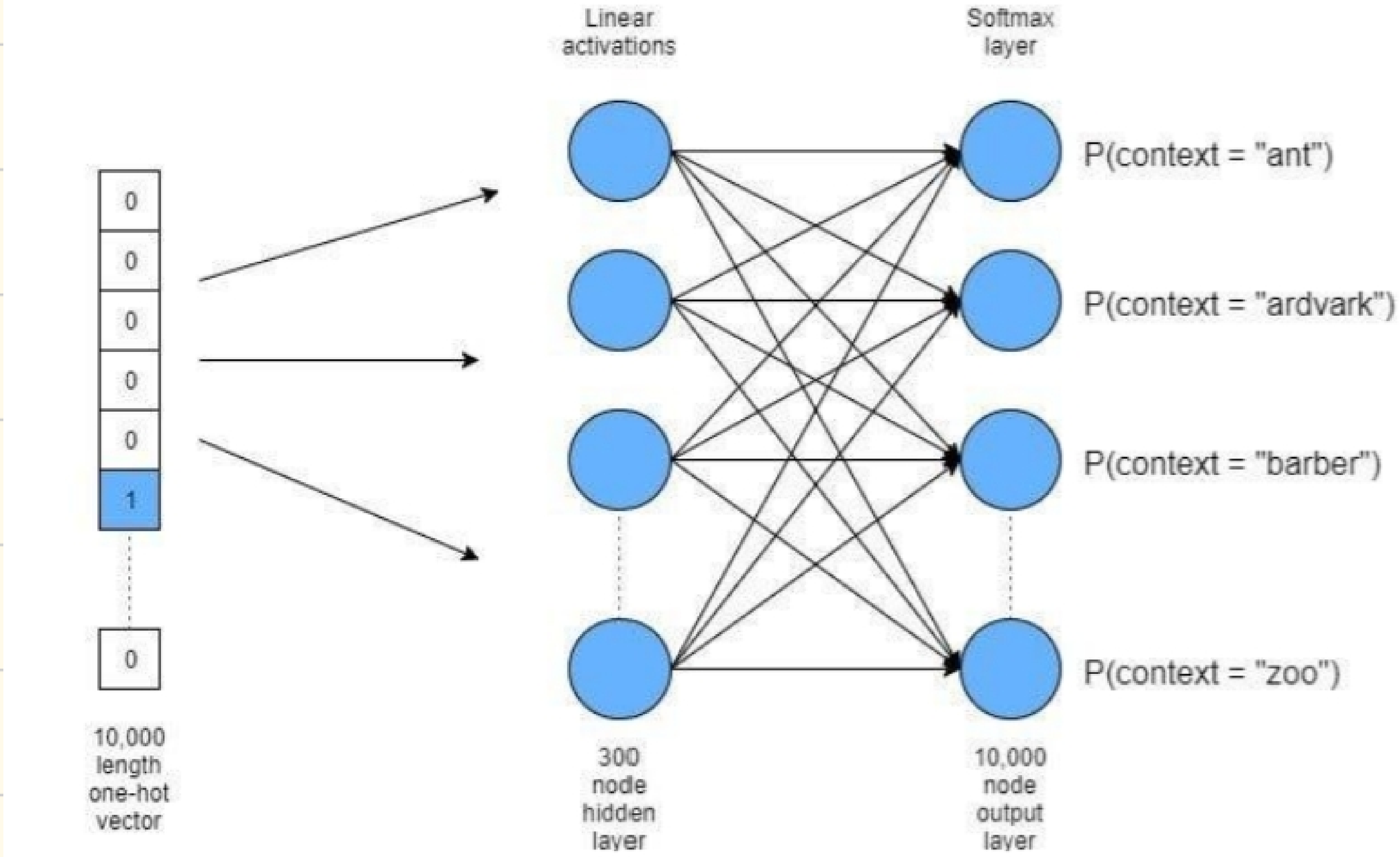
- predict 'office' and 'physics' based on 'science'.

- predict 'office' and 'science' based on 'physics'.

→ A loss function (encouraging prediction of similar words) would transform weights of each word.

→ These learned weights are stored as embeddings (Word2Vec like models).

→ Note that here only one vector would be there for a word, therefore can't distinguish b/w (river bank) and (financial bank) as two separate words.



So basically embedding vector shows which words it can predict - logically what is its semantic meaning.

